

1. Consider the [hospital](#) dataset at the blue link in this sentence. Fit a multiple regression model using `lm()`, with **infection_risk** as the response variable and all remaining **numerical** variables as the explanatory variables (do not include **region**).
 - (a) Plot the response against a numerical explanatory variable of your choice (You do not need to plot all of them).
 - (b) Simplify the model by removing ineffective explanatory variables, **one at a time**. Don't remove the intercept, and clearly identify your final reduced model.
 - (c) Check the assumptions of your reduced model, using **residual plots** and multiple sentences to explain.
 - (d) Interpret a **single** slope of your choice within the model in context of the data (you don't have to do all of them).
 - (e) Calculate 95% confidence intervals for all the coefficients and interpret **one** of them in context.
 - (f) Conclude an implicit hypothesis test for **one** non-intercept coefficient with the appropriate confidence interval. (In short, is the slope significantly different from zero or not?)

2. Consider the finches data set once more, [finches](#), found at the blue link. All units are in millimeters.
 - (a) Create a **first** linear model that predicts winglength based on beakwidth and island, using "beakwidth + island" in your code. List the code used to produce the model, as well as its output using the `summary()` function.
 - (b) Next, create a **second** linear model, but use "beakwidth + beakwidth:island" in the model instead. List the code and summary output for this model as well.
 - (c) To finish, create a **third** linear model using "beakwidth*island" in the model, listing your code and summary output.
 - (d) Based on the **first** model, what is the expected winglength of a bird from the island "floreana" with a beakwidth of 10.8 mm? Show your calculations.
 - (e) Based on the **second** model, what is the slope of the model for a bird from the island "santacruz"? Show your calculations and interpret this slope (e.g. "For every 1 unit increase...").
 - (f) What is the difference between the summary outputs for the **second** and **third** models?